HVPF-2: 80/20 Training and Trained Source Camera Identification

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Abstract

HVPF-2 is a copy of the HVPF pipeline with two changes. (1) The provenance models (organic / manipulated / synthetic) are trained on 80% of the camera models and tested on 20%, instead of 60/20/20; they are compared with the Phase 3 models on the same test sets. (2) Source camera identification is also learned: a Vision Transformer is trained to name the camera of a video (80/20 split of each camera's recordings) and compared with training-free PRNU matching and with PDC-ViT. Data, features and frames are identical to Phase 3.

Key findings

- **80/20 vs 60/20/20:** on the same test folds, single models gain +0.000 to +0.011 macro F1 (mean +0.006); the best model, stacking, stays at 0.850 (60/20/20: 0.851). The gains are smaller than the fold-to-fold spread (std 0.015): more training cameras help a little, not decisively. On unseen sources (FloreView + GenBuster, fixed split) stacking reaches 0.893 (0.863).
- **Trained camera identification:** the ViT names the camera of 90.6% (Video-ACID) and 92.7% (FloreView) of the test videos, against 58.4% and 26.8% for PRNU matching; on VISION it reaches only 47.7% (PRNU 58.0%).
- **Stabilised phones:** where PRNU fails (10.7% on Video-ACID, 8.5% on FloreView), the ViT still identifies 90.7% and 91.1%. For these phones the sensor pattern is unusable, so the ViT must rely on other traces of the camera, such as its image processing, stabilisation and encoder settings.
- **Model vs device:** on Video-ACID the ViT finds the right camera *model* for 99.6% of the videos but the right device for 90.6%: devices of the same model are what it confuses. Telling those apart is exactly what PRNU does when it is available.
- **Hybrid:** using PRNU when it is confident and the ViT otherwise gives the best accuracy on every dataset: 66.9% (VISION), 95.0% (Video-ACID), 93.8% (FloreView); PDC-ViT reports 94.3% and 74.77% on VISION and Video-ACID.
- **Why VISION is hard for the ViT:** about 30 training videos per camera, a split that keeps each scene on one side, and test videos that include WhatsApp and YouTube copies. Video-ACID has about 160 training videos per camera, and ACID and FloreView have no scene labels, so their splits are less strict (see Notes).

1 Provenance models: 80/20 vs 60/20/20

Both versions use the same camera-model groups. In 5-fold cross-validation, fold f is the test set in both; 60/20/20 trains on three folds and uses fold $f+1$ for validation, while 80/20 trains on all

four other folds. To keep model selection honest, 80/20 still uses fold $f+1$ to choose settings (SVM C, XGBoost depth and number of trees, the networks' number of epochs; calibration and stacking weights), then retrains on all four folds with those settings. The fixed split merges the old train and validation camera models (82% of the models) and keeps the same test models.

Table 1: Realistic setup, 5-fold cross-validation (main result), 5 frames: 60/20/20 (Phase 3) vs 80/20 (HVPF-2).

	Macro F1			PVA		AUC genuine vs not	
Model	60/20/20	80/20	change	60/20/20	80/20	60/20/20	80/20
PCE rule	–	–	–	–	–	0.549	0.549
SVM	0.769	0.780	+0.011	0.765	0.775	0.884	0.889
XGBoost	0.794	0.799	+0.005	0.791	0.794	0.913	0.915
GRU	0.779	0.782	+0.003	0.773	0.778	0.906	0.908
1D-CNN	0.774	0.782	+0.007	0.768	0.775	0.895	0.904
Transformer	0.780	0.780	+0.000	0.770	0.771	0.902	0.906
ViT (residual)	0.820	0.827	+0.007	0.806	0.814	0.912	0.916
Ens. all	0.814	0.818	+0.003	0.807	0.811	0.927	0.929
Ens. top-3 (val)	0.827	0.827	+0.001	0.818	0.820	0.932	0.935
Stacking (LR)	0.851	0.850	-0.001	0.836	0.838	0.941	0.941

Table 2: Encoder-controlled setup, 5-fold cross-validation, 5 frames: 60/20/20 (Phase 3) vs 80/20 (HVPF-2).

	Macro F1			PVA		AUC genuine vs not	
Model	60/20/20	80/20	change	60/20/20	80/20	60/20/20	80/20
PCE rule	–	–	–	–	–	0.513	0.513
SVM	0.751	0.757	+0.005	0.743	0.748	0.860	0.865
XGBoost	0.768	0.774	+0.005	0.759	0.763	0.886	0.889
GRU	0.756	0.767	+0.011	0.748	0.758	0.876	0.884
1D-CNN	0.758	0.761	+0.004	0.748	0.753	0.872	0.878
Transformer	0.756	0.756	-0.001	0.748	0.746	0.872	0.872
ViT (residual)	0.763	0.773	+0.010	0.753	0.763	0.856	0.863
Ens. all	0.785	0.792	+0.006	0.776	0.782	0.898	0.901
Ens. top-3 (val)	0.789	0.794	+0.005	0.780	0.783	0.897	0.900
Stacking (LR)	0.816	0.818	+0.002	0.805	0.807	0.906	0.909

Table 3: Realistic setup, fixed split, test cameras, 5 frames: 60/20/20 (Phase 3) vs 80/20 (HVPF-2).

	Macro F1			PVA		AUC genuine vs not	
Model	60/20/20	80/20	change	60/20/20	80/20	60/20/20	80/20
PCE rule	–	–	–	–	–	0.467	0.467
SVM	0.793	0.796	+0.003	0.796	0.800	0.898	0.904
XGBoost	0.806	0.811	+0.005	0.814	0.818	0.936	0.940
GRU	0.800	0.801	+0.001	0.797	0.805	0.933	0.933
1D-CNN	0.801	0.807	+0.006	0.801	0.804	0.930	0.930
Transformer	0.793	0.767	-0.026	0.789	0.779	0.927	0.927
ViT (residual)	0.834	0.849	+0.016	0.828	0.837	0.931	0.933
Ens. all	0.824	0.826	+0.003	0.824	0.827	0.946	0.946
Ens. top-3 (val)	0.841	0.836	-0.004	0.839	0.837	0.951	0.952
Stacking (LR)	0.866	0.874	+0.007	0.856	0.861	0.954	0.954

Table 4: Unseen sources (FloreView + GenBuster), fixed split, 5 frames: 60/20/20 (Phase 3) vs 80/20 (HVPF-2).

	Macro F1			PVA		AUC genuine vs not	
Model	60/20/20	80/20	change	60/20/20	80/20	60/20/20	80/20
PCE rule	–	–	–	–	–	0.406	0.406
SVM	0.630	0.649	+0.020	0.564	0.587	0.793	0.827
XGBoost	0.728	0.734	+0.005	0.641	0.651	0.922	0.931
GRU	0.619	0.714	+0.095	0.534	0.666	0.825	0.886
1D-CNN	0.660	0.603	-0.057	0.598	0.549	0.844	0.839
Transformer	0.715	0.668	-0.047	0.598	0.618	0.889	0.840
ViT (residual)	0.842	0.837	-0.005	0.815	0.808	0.942	0.956
Ens. all	0.722	0.723	+0.001	0.644	0.667	0.913	0.920
Ens. top-3 (val)	0.776	0.818	+0.041	0.696	0.756	0.944	0.961
Stacking (LR)	0.863	0.893	+0.030	0.818	0.847	0.965	0.978

2 Source camera identification by training

Each camera's recordings are split 80/20: 20% of its natural videos are test videos; all other videos, including the flat reference videos, are training videos. All versions of a recording (VISION native, YouTube and WhatsApp) stay on the same side, and VISION's moving, still and panning takes of one scene count as one recording, so a test scene is never seen in training (ACID and FloreView have no scene labels). 10% of the recordings are held out inside the training side to choose the number of epochs; the model is then retrained on the whole 80%. The model is the ViT-S/16 of Step 8b with one output per camera, on SRM noise residuals. *PRNU* is the training-free baseline of Step 4 on the same test videos (fingerprints from the reference videos, which are never test videos). *Hybrid* uses PRNU when it verifies the camera ($PCE_0 > 60$) and the ViT otherwise; *hybrid (val)* chooses that PCE_0 threshold on the validation videos.

Table 5: Top-1 camera identification accuracy (%) on the test videos. Model level: a prediction of another camera of the same model counts as correct.

Dataset	Test videos	Cameras	ViT	PRNU	Hybrid	Hybrid (val)	ViT model level
VISION	350	35	47.7	58.0	60.3	66.9	53.1
usable PRNU	214	21	55.6	84.6	74.8	84.1	57.9
no usable PRNU	136	14	35.3	16.2	37.5	39.7	45.6
version native	118	35	56.8	72.0	83.1	84.7	67.8
version youtube	114	35	49.1	61.4	57.0	67.5	51.8
version whatsapp	118	35	37.3	40.7	40.7	48.3	39.8
Video-ACID	1756	44	90.6	58.4	94.1	95.0	99.6
usable PRNU	996	25	90.6	94.8	96.7	98.4	99.4
no usable PRNU	760	19	90.7	10.7	90.7	90.5	99.9
FloreView	385	45	92.7	26.8	93.2	93.8	92.7
usable PRNU	80	10	98.8	96.2	100.0	100.0	98.8
no usable PRNU	305	35	91.1	8.5	91.5	92.1	91.1

2.1 Comparison with PDC-ViT

PDC-ViT (Elharrouss et al. 2025) also trains a Vision Transformer with an 80/20 split; its numbers are taken from the paper (Video-ACID there includes WhatsApp copies, 26 cameras). FNR and FPR are macro averages over the cameras (one camera vs the rest).

Table 6: Camera identification compared with PDC-ViT.

Method	Dataset (cameras)	Accuracy (%)	FNR (%)	FPR (%)
PDC-ViT (reported)	VISION (35)	94.30	–	–
PDC-ViT (reported)	Video-ACID + WA (26)	74.77	23.4	0.92
HVPF-2 ViT	VISION (35)	47.71	55.6	1.54
PRNU	VISION (35)	58.00	44.2	1.24
Hybrid (val)	VISION (35)	66.86	36.6	0.97
HVPF-2 ViT	Video-ACID (44)	90.60	9.5	0.22
PRNU	Video-ACID (44)	58.37	41.5	0.97
Hybrid (val)	Video-ACID (44)	94.99	5.0	0.12
HVPF-2 ViT	FloreView (45)	92.73	7.0	0.17
PRNU	FloreView (45)	26.75	71.2	1.66
Hybrid (val)	FloreView (45)	93.77	6.0	0.14

3 Notes

- The provenance test sets are identical in both versions, so the differences come only from training on 80% instead of 60% of the camera models (and from the refit on all of it).
- Camera identification is closed-set: every test camera is in training. Provenance stays open-set: test camera models are never seen in training.
- ACID and FloreView have no scene labels, so two videos of the same scene by one camera may fall on different sides of the split; the SRM residual input makes scene content less useful to the model, but not useless.